

B.Tech. DEGREE EXAMINATION, FEB 2023.

First Semester

MATHEMATICS – I

(Common to All branches)

(From 2013-14 onwards)

Time : Three hours

Maximum : 75 marks

Date:

PART A – (10 × 2 = 20 marks)

Answer ALL Questions

All questions carry equal mark

1. Define curvature and radius of curvature.
2. What is the relation between gamma and beta function.
3. If $u = \frac{x}{y} + \frac{y}{z}$, find $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z}$.
4. Expand $e^x \log(1+y)$ in powers of x and y upto terms of second degree.
5. Evaluate $\int_0^1 \int_2^5 xy \, dx \, dy$
6. Evaluate $\int_0^a \int_0^b \int_0^c xyz \, dx \, dy \, dz$
7. Solve $(y \cos x + 1)dx + \sin x \, dy = 0$.
8. Solve $(y - px)(p - 1) = p$.
9. Find the P.I of $\frac{d^2 y}{dx^2} + y = \sin x$.
10. Solve: $(D^2 + 3D + 2)y = 0$.

PART B – (5 × 11 = 55 marks)

Answer ALL the Units, choosing ONE full question from each unit

UNIT - I

11.a) Find the evolute of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

11. b) Find the radius of curvature at the point $(a \cos^3 \theta, a \sin^3 \theta)$ on the curve $x^{2/3} + y^{2/3} = a^{2/3}$.

(OR)

12.a) State and prove the relation between beta and gamma functions.

12. b) Prove that the evolute of the rectangular hyperbola $xy = c^2$.

UNIT - II

13. a) A rectangular box open at the top, is to have a volume of 32cc. Find the dimensions of the box, the requires the least material for its construction.

13.b) Find the extreme values of $f(x,y) = x^3 + y^3 - 3x - 12y + 20$

(OR)

14.a) If $u = x^2 + y^2 + z^2$, $v = xy + yz + zx$, $w = x + y + z$. Find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$.

14.b) Find the Taylors series expansion upto second degree for

$f(x,y) = x^2 y + 3y - z$ at $(1,-2)$.

UNIT – III

15.a) Find the area enclosed between the curve

$$x^2 = 4ay \text{ and } y^2 = 4ax.$$

15.b) Find the volume bounded by cylinder $x^2 + y^2 = 4$ and the planes $y + z = 4$ and $z = 0$.

(OR)

16. Find the volume of the sphere $x^2 + y^2 + z^2 = a^2$.

UNIT – IV

17.a) Solve: $y = 2px + y^2p^3$.

$$17.b) \frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} = 0$$

(OR)

18.a) Solve $p^2 + 2py \cot x = y^2$.

18.b) Find the orthogonal trajectories of the family of the parabola $y^2 = 4ax$

UNIT – V

19. Solve $(D^2+1)y = \operatorname{cosec} x$ using method of variation of parameter.

(OR)

20. $(D^3 - 3D^2 + 4D - 2)y = e^x + \cos x$.